

Harmonic-coordinate worldtube matching

Analytic structure and numerical findings from
 $q = 4$ and $q = 8$ virtual worldtubes

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Purpose. This report is the interpretive companion to `q8_first_order_matching.ipynb`. The notebook contains the executable block solves, manufactured tests, closure diagnostics, radial derivatives, random-point validation, second-order sensitivity analysis, and Gauss–Seidel iteration. The focused directory `temporal_ode_consistency/` contains the multi-slice and direct Π, Φ time-derivative checks.

Abstract

We investigate whether a tidally perturbed Schwarzschild metric in harmonic coordinates can supply inner-boundary data from numerical binary-black-hole simulations. Expanding the coordinate map and inverse metric through second order turns the matching problem into a hierarchy of linear overdetermined systems once lower-order sources are fixed. Component, parity, and STF structure split those systems into small blocks whose coefficients can be cross-checked in omitted multipoles, radial derivatives, and independent interior points.

The first-order model gives a large and repeatable improvement over centered Schwarzschild. At $q = 8$, the provisional V1+C3 hierarchy reduces the held-out inverse-metric RMS error from 2.21×10^{-1} to 3.61×10^{-2} . Radial derivatives modestly improve the vector sector. The complete triangular 43-parameter second-order solve is algebraically correct but fails cross-component and interior validation. Sensitivity is concentrated in the symmetric spatial-strain rate and time acceleration, while a targeted polar- $J = 2$ subsystem provides the strongest larger second-order block. Early-stopped block Gauss–Seidel iteration reduces the interior error to approximately 3.0×10^{-2} on the smallest worldtubes.

Direct metric time derivatives constructed from Π, Φ , lapse, and shift show that freely fitted second-order time jets were inflated by roughly three orders of magnitude. They also provide independent measurements of first-order evolution and, through the fixed-inertial translation response, a metric-only measurement of the center velocity.

Contents

1	Scientific objective and scope	3
1.1	Status vocabulary	3
2	Coordinate map and perturbative organization	4
3	Response functions and the linear hierarchy	4
3.1	Where nonlinearity remains	5
4	Fixed-final-point response and symbolic audit	5

5	Center measurement	6
5.1	Kretschmann dipole	6
5.2	Metric-only translation response	6
6	First-order block decomposition	6
6.1	Worked block example	7
7	Closure and held-out validation	7
8	First-order numerical findings	7
8.1	q4 baseline	7
8.2	q8 provisional hierarchy	8
9	Radius dependence and sensitivity	8
10	Spatial derivatives	11
10.1	Derivative as held-out closure	11
10.2	Joint value-and-derivative fit	11
11	Second-order decomposition	11
12	Why the complete triangular solve fails	12
13	Recoverable second-order information	12
13.1	Polar- $J = 2$ subsystem	13
14	Second-order radial derivatives	13
15	Block Gauss–Seidel decontamination	14
16	Time derivatives and evolution consistency	15
16.1	First-order evolution and center velocity	16
16.2	Metric-only translation velocity	17
17	Limitations	18
18	Current assessment	18
A	Compact numerical summary	19
B	Companion artifacts	19

1 Scientific objective and scope

The purpose of the worldtube construction is to replace the unresolved interior of the smaller black hole by a local analytic metric whose coefficients are measured from the numerical solution on a surrounding sphere. The local metric is a slowly evolving, tidally perturbed Schwarzschild solution in harmonic coordinates.

The fit must accomplish more than interpolation on the fitting sphere. A useful matching model must:

1. determine the coordinate transformation between the numerical inertial coordinates and local harmonic coordinates;
2. measure the physical tidal fields and coordinate-map coefficients;
3. predict metric component/STF blocks omitted from the solve;
4. reproduce independent spatial and time derivatives;
5. extrapolate the inverse metric to points inside the worldtube.

Two simulations have been used:

- a $q = 4$ calculation, principally at $T = 850M_{\text{tot}}$, for the initial analytic and numerical audit;
- a more perturbative $q = 8$ calculation at $T = 950M_{\text{tot}}$, together with ten slices through $T = 995M_{\text{tot}}$, for the current block, derivative, and dynamical tests.

The black-hole mass M is fixed to its initial-data value. It is not a fitting parameter, and tidal heating enters beyond the retained order. The formal parameter ϵ counts perturbative order. It is set to one when a complete truncated metric is evaluated; it is not a numerical velocity.

1.1 Status vocabulary

Status	Meaning
algebraically verified	Independent symbolic constructions agree, or a manufactured identity vanishes to roundoff.
numerically verified	A stated claim has passed a simulation-data test with a recorded diagnostic.
provisional	A trend is useful for designing the matching system but has not yet been demonstrated across enough simulations, times, or resolutions.
open	The present analysis does not establish the claim.

2 Coordinate map and perturbative organization

Let $(t, \rho n^i)$ be local harmonic coordinates and (T, X^i) the simulation inertial coordinates. At fixed harmonic slow time, the coordinate map through second order is

$$T = t + q^0 + \epsilon \frac{\rho}{M} b_i n^i + \epsilon^2 \left[\mathcal{J}(\rho) \dot{q}^0 + \mathcal{K}(\rho) \dot{b}_i n^i + \Phi(\rho) Q_{ij} n^i n^j - M(\rho^2 - M^2) \mathcal{E}_{ij} n^i n^j \right] + O(\epsilon^3), \quad (1)$$

$$X^i = \rho n^i + q^i + \epsilon \frac{\rho}{M} L^i_j n^j + \epsilon^2 \left[\mathcal{J}(\rho) \dot{q}^i + \mathcal{K}(\rho) \dot{L}^i_j n^j + \Phi(\rho) Q^i_{jk} n^j n^k + \text{electric-tidal radial profiles} \right] + O(\epsilon^3). \quad (2)$$

Here

$$\mathcal{J}(\rho) = W_D(\rho) - \frac{1}{2}(\rho_D^*)^2, \quad \mathcal{K}(\rho) = -2(\rho - 2M), \quad \Phi(\rho) = \frac{1}{2} \left[3(\rho/M)^2 - 1 \right], \quad (3)$$

with $\rho_D^* = r + 2M \ln[r/(2M)]$ and $r = \rho + M$. The explicit electric profiles and the complete light-cone metric are recorded in the analytic audit package; the present report concentrates on how their response columns are used numerically.

Order	quantity	count	role
ϵ^0	q^i, q^0	3 + 1	spatial center and conventional time origin
ϵ^1	$\dot{q}^0, b_i, \dot{q}^i, L_{(ij)}$	1 + 3 + 3 + 6	clock rate, time gradient, translational velocity, and symmetric affine map
ϵ^2	$\ddot{q}^\mu, \dot{b}_i, \dot{L}_{(ij)}$	4 + 3 + 6	instantaneous time jets; evolution variables across slices
ϵ^2	Q_{ij}, Q^i_{jk}	5 + 15	new harmonic-coordinate quadrupoles
ϵ^2	$\mathcal{E}_{ij}, \mathcal{B}_{ij}$	5 + 5	physical electric and magnetic STF tensors

The local triad is taken to be nonrotating, so the antisymmetric part of L^i_j is fixed by convention. Inertial orbital motion remains encoded in $q^i(t)$, $\dot{q}^i$, b_i , and the symmetric map.

3 Response functions and the linear hierarchy

Expand the inverse metric as

$$G^{ab} = G_{(0)}^{ab} + \epsilon G_{(1)}^{ab} + \epsilon^2 G_{(2)}^{ab} + O(\epsilon^3). \quad (4)$$

At order n ,

$$G_{(n)}^{ab} = S_{(n)}^{ab}[p_{(0)}, \dots, p_{(n-1)}] + \sum_A \mathcal{R}_{(n)A}^{ab} p_{(n)}^A. \quad (5)$$

The source $S_{(n)}$ has total order n , even though it is constructed from lower-order variables. For example, $p_{(1)} p_{(1)}$ is part of the second-order source.

A response function is

$$\mathcal{R}_A^{ab}(x) = \frac{\partial G^{ab}(x; p)}{\partial p^A}. \quad (6)$$

After projecting into a metric component and STF block B , the matching equation becomes

$$d_B = A_{BA} p^A + e_B. \quad (7)$$

Here e_B contains numerical error, truncation error, and all terms missing from the current model. The exact response is block diagonal under rotation and parity, but finite-order contamination need not respect the intended separation between perturbative orders.

3.1 Where nonlinearity remains

The order-by-order systems are linear only after lower-order variables are fixed. Four effects require additional care:

1. The spatial center changes the background evaluation point and is therefore found by linearized translation updates and iteration.
2. At second order, the known source $Q_{11}[p_{(1)}, p_{(1)}]$ is quadratic in first-order coefficients.
3. Across time, $\ddot{q} = d\dot{q}/dt$, $\dot{b} = db/dt$, and $\dot{L} = dL/dt$ make the full system differential-algebraic.
4. At finite mass ratio, unresolved higher-order structure can project onto lower-order response columns. A well-conditioned least-squares solution can therefore still be physically wrong.

4 Fixed-final-point response and symbolic audit

The numerical metric is evaluated at fixed final inertial points. For a coordinate displacement ξ^i , the inverse-metric response is

$$\delta G^{\mu\nu} = (\partial\xi)^\mu{}_\alpha G_{(0)}^{\alpha\nu} + (\partial\xi)^\nu{}_\alpha G_{(0)}^{\mu\alpha} - \xi^k \partial_k G_{(0)}^{\mu\nu}. \quad (8)$$

The last term transports the spatially varying Schwarzschild background. It was missing from an early spatial-strain implementation. Finite differentiation of the exact inverse point map and complete metric pushforward verifies the corrected response to better than 4×10^{-9} .

Two independent Mathematica calculations were used at second order:

1. a direct Dhesis-style coefficient solve for the inverse map;
2. recursive inversion of the forward map followed by the metric pushforward at fixed final coordinates.

The implemented b^2 , bL , L^2 , $\dot{L}$, electric, temporal/spatial Q , and $\ddot{q}^\mu$ families agree algebraically. Both compositions give the identity through $O(\epsilon^2)$. Exact nonzero composition residuals were found in printed quadratic- L and electric inverse rules from the thesis; the corrected direct and recursive constructions remove them. The complete magnetic sign and normalization remain **open**.

5 Center measurement

5.1 Kretschmann dipole

For Schwarzschild,

$$\mathcal{K} = \frac{48M^2}{r^6}, \quad \rho\mathcal{K} = \left(\frac{48M^2}{\mathcal{K}}\right)^{1/6} - M. \quad (9)$$

On a sphere of radius R , define

$$F(n) = R^2 - \rho_{\mathcal{K}}^2. \quad (10)$$

For a small displacement δz^i ,

$$F(n) = 2R\delta z_i n^i + \dots, \quad (11)$$

so the Cartesian dipole supplies a Newton correction. The numerical sphere is reinterpolated after each correction.

On the q4 slice, an injected displacement of $0.01M_{\text{tot}}$ was recovered with a finite-radius error of about 12%, while the outer iteration converged to updates near $10^{-10}M_{\text{tot}}$. The converged center retained a radius drift near $10^{-3}M_{\text{tot}}$, consistent with finite-radius and higher-order effects.

On q8 at $T = 950M_{\text{tot}}$, the curvature center differs from the apparent-horizon center by

$$(1.77 \times 10^{-4}, 7.44 \times 10^{-5}, 0)M_{\text{tot}}, \quad (12)$$

and the final update at $R = 0.2M_{\text{tot}}$ is approximately $5 \times 10^{-13}M_{\text{tot}}$.

The curvature center is an operational expansion center. It is not assumed to coincide exactly with the apparent-horizon areal center or a physical singularity position.

5.2 Metric-only translation response

A translated Schwarzschild metric obeys

$$G_{(0)}^{ab}(x - \delta z) = G_{(0)}^{ab}(x) - \delta z^k \partial_k G_{(0)}^{ab}(x) + O(\delta z^2). \quad (13)$$

The translation columns are therefore

$$C_k^{ab} = -\partial_k G_{(0)}^{ab}. \quad (14)$$

They appear in G_1^{TT} , $G_{0,2}^{Ti}$, and $G_{1,3}^{ij}$. Within the present first-order model, G_3^{ij} is the cleanest translation-only block. The remaining blocks overlap with the time-gradient and velocity responses.

6 First-order block decomposition

The thirteen first-order parameters are

$$p_{(1)} = (\dot{q}^0, b_i, \dot{q}^i, \sigma_{ij}), \quad \sigma_{ij} = L_{(ij)}/M. \quad (15)$$

Parity and STF structure separate them into the vector and clock/strain sectors:

sector	unknowns	candidate blocks
vector	$b_i, \dot{q}^i$	$G_1^{TT}, G_{0,2}^{Ti}, G_1^{ij}$
clock/strain	$\dot{q}^0, \sigma_{ij}$	$G_{0,2}^{TT}, G_{1,3}^{Ti}, G_{0,2,4}^{ij}$

The five inclusion-minimal full-rank estimators in each sector are

$$\begin{aligned} V1 : G_1^{TT} + G_0^{Ti}, & & V2 : G_1^{TT} + G_1^{ij}, \\ V3 : G_0^{Ti} + G_2^{Ti}, & & V4 : G_0^{Ti} + G_1^{ij}, \\ V5 : G_2^{Ti} + G_1^{ij}, & & \end{aligned}$$

and

$$\begin{aligned} C1 : G_0^{TT} + G_1^{Ti}, & & C2 : G_0^{TT} + G_0^{ij}, \\ C3 : G_0^{TT} + G_2^{ij}, & & C4 : G_1^{Ti} + G_0^{ij}, \\ C5 : G_1^{Ti} + G_2^{ij}. & & \end{aligned}$$

6.1 Worked block example

For one Cartesian direction, G_1^{TT} responds primarily to b_i , whereas G_0^{Ti} contains an independent combination of b_i and $\dot{q}^i$. V1 therefore gives a six-parameter full-rank vector solve. The recovered coefficients then predict G_2^{Ti} and G_1^{ij} , neither of which entered the fit.

Manufactured data recover all ten minimal estimators to better than 10^{-14} , with omitted-block closure below 10^{-13} . Disagreement on simulation data is therefore not caused by column ordering or rank selection.

7 Closure and held-out validation

For parameters inferred from fitted blocks, the fractional omitted-block closure is

$$\mathcal{C} = \frac{\|d_{\text{omitted}} - A_{\text{omitted}}\hat{p}\|_2}{\|d_{\text{omitted}}\|_2}. \quad (16)$$

Fractional and absolute RMS residuals are always retained together. A radially weak physical block can have a large fractional residual even while its absolute mismatch decreases.

Four increasingly strong diagnostics are used:

1. residuals in blocks included in the solve;
2. component/STF blocks omitted from the solve;
3. radial derivatives from the evolved Φ_{iab} ;
4. the complete inverse metric at random points inside the worldtube.

The random-point validation is the primary out-of-sample test. At q8, 768 points occupy three volume-uniform shells inside $R = 0.2M_{\text{tot}}$. They never enter a surface or derivative solve.

8 First-order numerical findings

8.1 q4 baseline

At q4, different full-rank first-order estimators did not approach a unique coefficient vector. This ambiguity propagated into the interior metric:

model	best radius	relative RMS
centered Schwarzschild	–	2.17×10^{-1}
selected first order	$0.60M_{\text{tot}}$	4.44×10^{-2}
selected second order	$0.50M_{\text{tot}}$	3.80×10^{-2}

At small radii, the spread among viable first-order estimators was several times larger than the useful second-order correction. This motivated the larger-mass-ratio q8 test.

8.2 q8 provisional hierarchy

The q8 analysis uses $M_B = 1/9$, Spherpak $\ell_{\text{max}} = 16$, and $0.2M_{\text{tot}} \leq R \leq 1.6M_{\text{tot}}$. The provisional hierarchy is

$$V1: \quad G_1^{TT} + G_0^{Ti}, \quad C3: \quad G_0^{TT} + G_2^{ij}. \quad (17)$$

The main omitted vector closure is $G_2^{Ti} + G_1^{ij}$, while G_1^{Ti} is the principal clock/strain closure. The spatial monopole G_0^{ij} is retained as an absolute diagnostic but is not used to determine the small-radius clock/strain parameters.

The held-out random-point error falls from

$$E_0 = 2.21 \times 10^{-1} \quad (18)$$

for centered Schwarzschild to

$$E_1 = 3.61 \times 10^{-2} \quad (19)$$

for V1+C3 at $R = 0.2M_{\text{tot}}$, an improvement by a factor of 6.14.

9 Radius dependence and sensitivity

Many residuals are convex functions of worldtube radius: they initially decrease, reach a minimum, and then increase.

The large-radius branch is expected because the local expansion is evaluated farther from its center and several parameter-separating responses decay. The small-radius branch was tested more carefully. In harmonic Schwarzschild coordinates, response denominators contain $r = \rho + M$, not ρ , so the first-order clock, boost, velocity, and most strain responses remain finite as $\rho \rightarrow 0$.

The numerical Jacobian confirms that there is no small-radius explosion:

- raw response norms vary only by order-unity factors;
- minimal-estimator singular values generally improve inward;
- surface-to-interior amplification does not diverge.

The small-radius ambiguity instead localizes to clock/strain fits containing G_0^{ij} . That monopole retains an $O(1)$ strain response inward, so unmodeled higher-order monopole content can be absorbed into the trace strain and clock rate without causing a poor condition number.

Four distinct sensitivities were recorded:

1. response norm per unit coefficient;

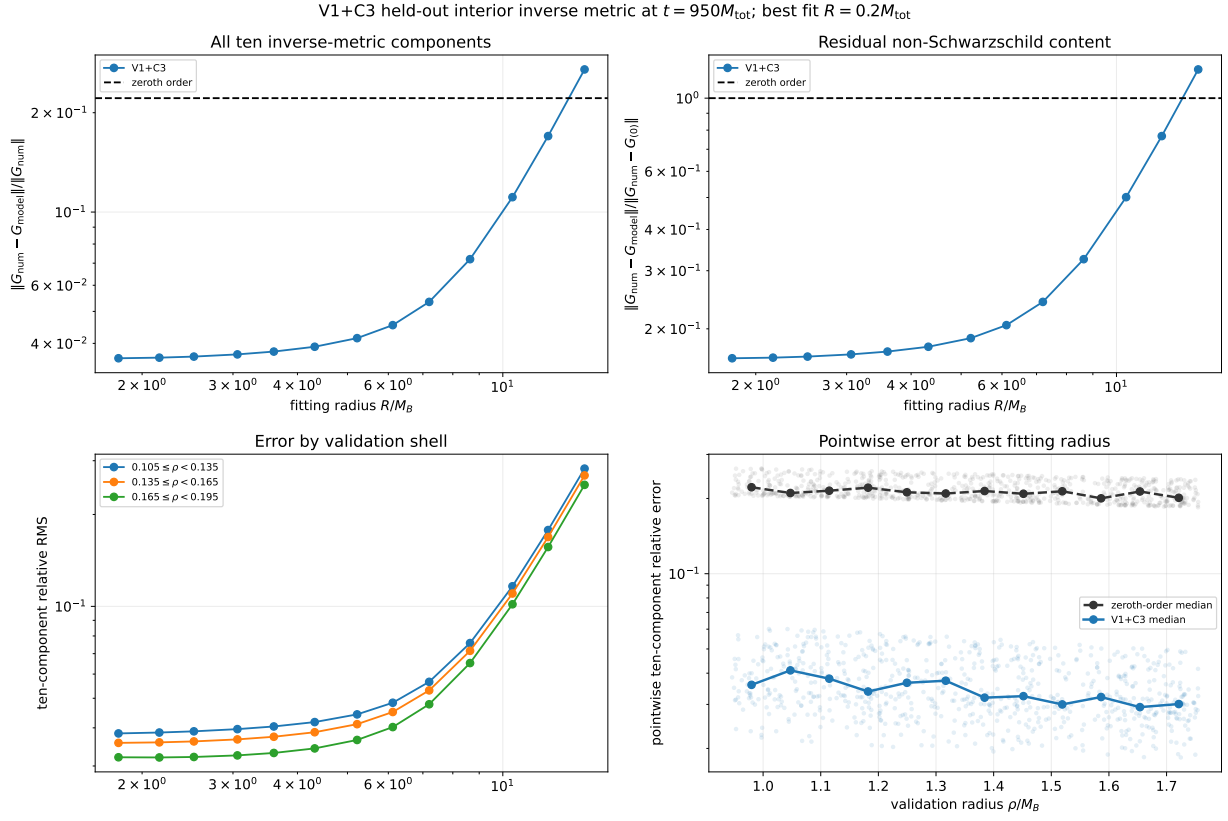


Figure 1: Held-out q8 inverse-metric prediction for the provisional V1+C3 hierarchy. The points are strictly inside the smallest fitting sphere and are not used by any surface fit.

2. smallest singular value of a block;
3. spread among viable estimators;
4. effect of that spread on the interior metric.

For $d_R = A_R p + h_R$,

$$\delta p = A_R^+ h_R, \quad \delta G_{\text{int}} = J_{\text{int}} A_R^+ h_R. \quad (20)$$

The norm of $J_{\text{int}} A_R^+$ measures surface-to-interior amplification. At small q8 radii, uncertainty in $\dot{q}^0$ has the largest individual interior impact. At large radius, orbital-plane components of $\dot{q}^i$ become dominant as the separating blocks decay.

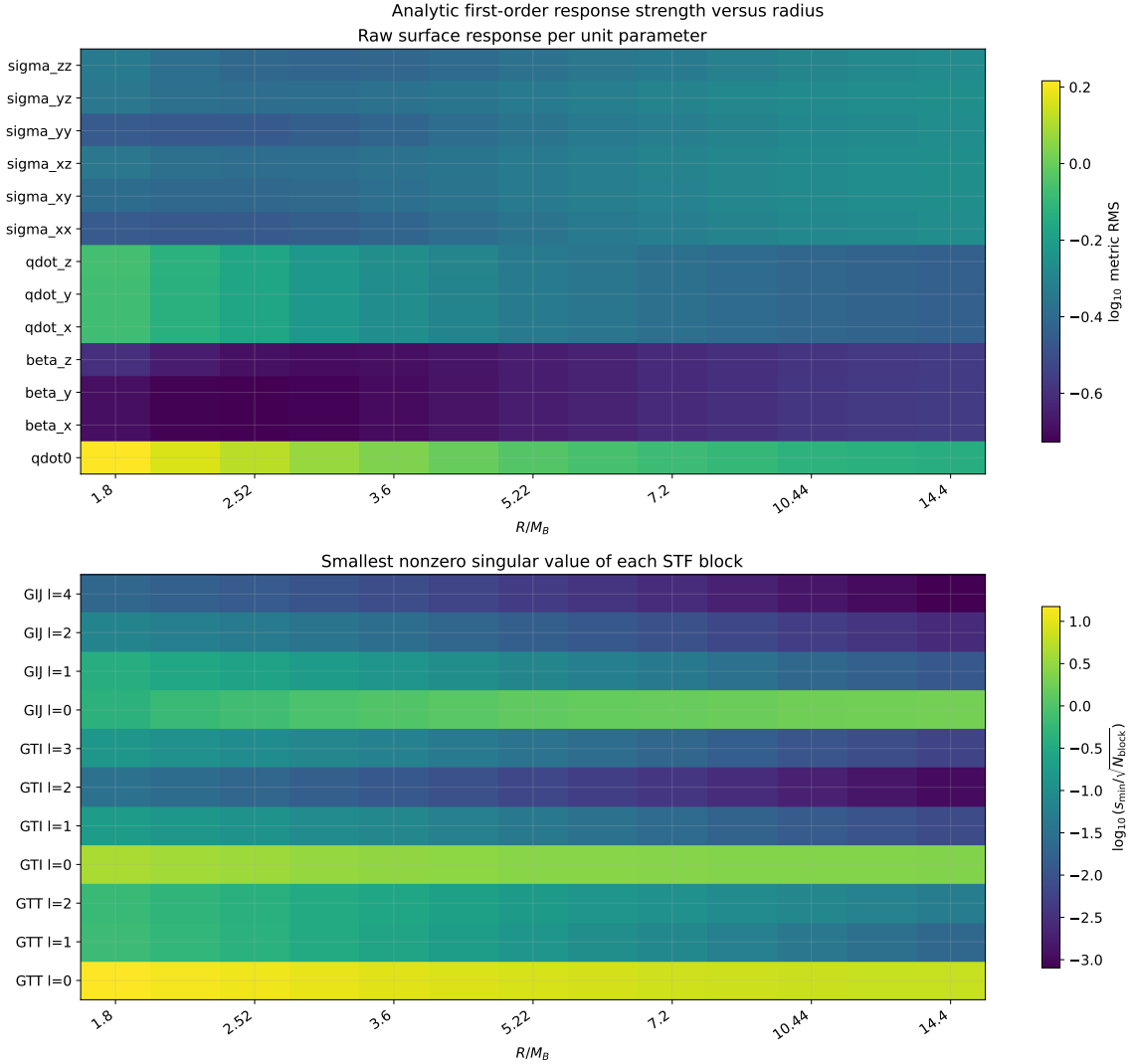


Figure 2: Radius-dependent first-order response strength, identifiability, and surface-to-interior amplification. The small-radius error branch is not caused by a divergent first-order response.

10 Spatial derivatives

The generalized-harmonic first-order reduction supplies

$$\Phi_{iab} = \partial_i g_{ab}, \quad (21)$$

so

$$\partial_i G^{ab} = -G^{ac} G^{bd} \Phi_{icd} \quad (22)$$

is available algebraically. The dimensionless radial observable is

$$\mathcal{D}G^{ab} = R n^i \partial_i G^{ab}. \quad (23)$$

The Φ -constructed result agrees with independent radial interpolation differences at 10^{-8} – 10^{-6} . Analytic response derivatives are converged to approximately 10^{-11} .

10.1 Derivative as held-out closure

Parameters fitted only to metric values predict the radial derivative with the following representative q8 residuals at $R = 0.2M_{\text{tot}}$:

- 2.1% in the V1 fitted blocks;
- 12% in the C3 fitted blocks;
- 14.6% in the omitted vector blocks.

10.2 Joint value-and-derivative fit

The augmented system is

$$\begin{pmatrix} A \\ w_D A_D \end{pmatrix} p \simeq \begin{pmatrix} d \\ w_D d_D \end{pmatrix}. \quad (24)$$

No new unknown is introduced. The best scanned first-order combination uses

$$w_D^{\text{vector}} = 3, \quad w_D^{\text{clock}} = 0. \quad (25)$$

It improves the random-point error from 0.036055 to 0.035699. The gain is modest but is supported by omitted vector value and derivative blocks. The C3 derivative currently degrades clock/strain closure and is therefore retained only as a diagnostic.

11 Second-order decomposition

After fixing the first-order baseline, the candidate residual is

$$r_{(2)} = G_{\text{num}} - G_{(0)} - A_1 p_{(1)} - Q_{11}[p_{(1)}, p_{(1)}]. \quad (26)$$

The 43 new/evolved coefficients are:

group	count
spatial acceleration $\ddot{q}^i$	3
symmetric spatial-strain rate $\dot{L}_{(ij)}$	6
electric STF tensor $\mathcal{E}_{ij}$	5
spatial quadrupoles $Q^i_{\langle jk \rangle}$	15
time acceleration $\ddot{q}^0$	1
time-gradient rate $\dot{b}_i$	3
temporal quadrupole $Q^0_{\langle ij \rangle}$	5
magnetic STF tensor $\mathcal{B}_{ij}$	5

The nominal triangular solve first fits 29 spatial coefficients in G^{ij} , then nine temporal-even coefficients in G^{TT} , and finally five magnetic coefficients in G^{Ti} . Manufactured tests recover all 43 parameters to approximately 8×10^{-12} , with metric closure near 10^{-14} .

12 Why the complete triangular solve fails

On q8 simulation data, the triangular system closes its determining blocks but fails to predict other components:

- the temporal G^{TT} block closes at 10^{-3} or better;
- the resulting G^{Ti} contribution is not supported by the numerical residual;
- the magnetic fit removes essentially none of that mismatch;
- the held-out interior error worsens to 0.097–0.101, compared with 0.0357–0.0361 at first order.

This is not a rank failure. It is a cross-component closure failure: the triangular estimator absorbs unmodeled content in the components that determine each stage.

Allowing a free nine-component correction to the first-order spatial variables is worse. The condition number grows to $O(10^4)$, corrections become comparable to the original first-order state, and the interior error increases to approximately 0.11.

13 Recoverable second-order information

Each physical group was fitted independently and tested by surface reduction, interior prediction, cross-block closure, and ablation.

- Spatial-strain rate lowers the best interior error by about 7.8%.
- Time acceleration lowers it by about 7.1%.
- Fitting those seven coefficients jointly lowers it by 10.6% for metric-only first order and by as much as 15.5% for the derivative-assisted baseline.
- Electric STF, temporal quadrupole, spatial acceleration, time-gradient rate, and spatial quadrupoles individually change the best error by less than about one percent.
- The magnetic response is numerically irrelevant at the present accuracy.

13.1 Polar- $J = 2$ subsystem

The fifteen-parameter polar block combines

$$G_2^{TT}, \quad G_3^{Ti}, \quad G_4^{ij} \quad (27)$$

to determine strain rate, electric STF, and temporal quadrupole coefficients. At the smallest q8 radius, the held-out G_3^{Ti} and G_4^{ij} residuals are about 10% and 5%, respectively. Its random-point error reaches 3.13×10^{-2} , compared with 3.57×10^{-2} for the best first-order baseline.

Spatial $J = 1$ and $J = 3$ quadrupole systems do not pass their held-out multipole tests and should not yet be interpreted as individual physical measurements.

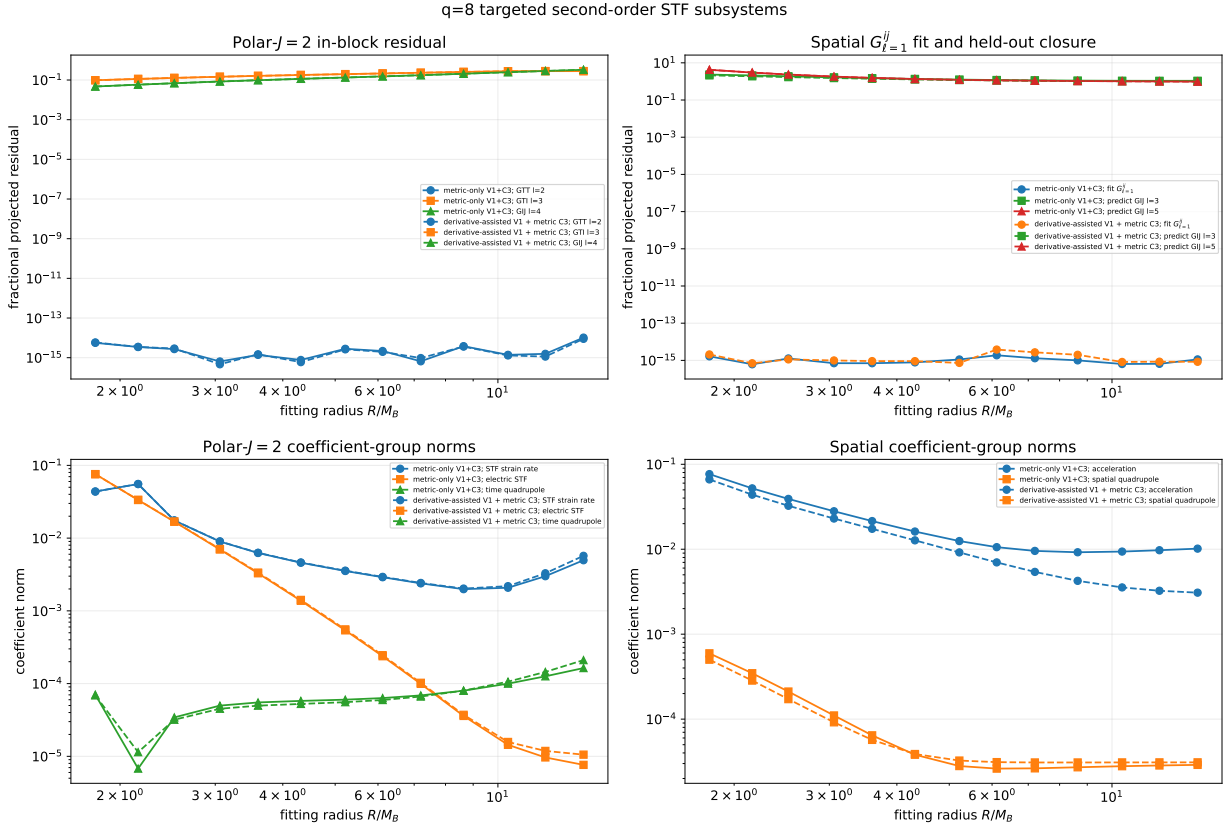


Figure 3: Targeted second-order STF systems. A fitted-block residual is not sufficient: the coefficients must predict other components and multipoles.

A globally balanced 43-column fit reaches an interior error near 3.01×10^{-2} , demonstrating useful aggregate second-order directions. It does not establish that all 43 coefficients are individually resolved. The conservative provisional subsystem is therefore $\dot{L}_{(ij)} + \dot{q}^0$.

14 Second-order radial derivatives

Adding $\mathcal{D}G^{ab}$ to the seven-parameter system with $w_D = 1$ reduces the best interior error from approximately 3.27×10^{-2} to 3.19×10^{-2} . With derivative-assisted first order, it falls from 3.24×10^{-2} to 3.17×10^{-2} .

The polar- $J = 2$ system benefits strongly from a small derivative weight: $w_D = 0.1$ reduces its poorly extrapolating value-only fit from roughly 4.1×10^{-2} to $3.32\text{--}3.35 \times 10^{-2}$. Larger weights improve the condition number but not the interior metric. Weight selection must therefore be based on held-out physical prediction rather than fitted residual or conditioning alone.

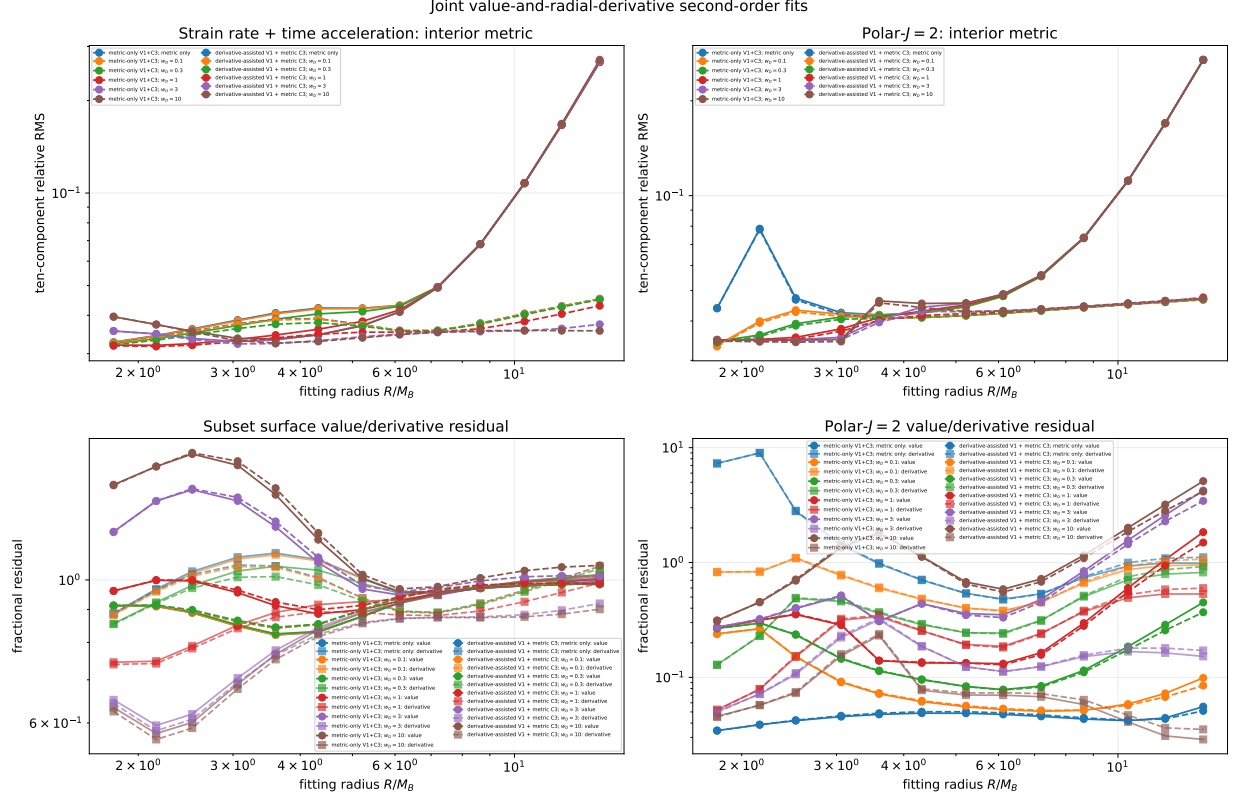


Figure 4: Joint value-and-radial-derivative fits for the seven-parameter and polar- $J = 2$ second-order systems.

15 Block Gauss–Seidel decontamination

Cross-order leakage motivates an alternating solve. At one radius,

$$d = A_1 p_1 + Q_{11}(p_1, p_1) + A_2 p_2, \quad (28)$$

$$d_D = A_{1D} p_1 + Q_{11,D}(p_1, p_1) + A_{2D} p_2. \quad (29)$$

Holding $p_1^{(k)}$ fixed,

$$p_{2,\star}^{(k+1)} = F_2 \left[d - A_1 p_1^{(k)} - Q_{11}(p_1^{(k)}, p_1^{(k)}) \right]. \quad (30)$$

The first-order update then reuses the validated V1+C3 estimators:

$$p_{1,\star}^{(k+1)} = F_1 \left[d - Q_{11}(p_1^{(k)}, p_1^{(k)}) - A_2 p_2^{(k+1)} \right]. \quad (31)$$

Relaxation is applied through

$$p_j^{(k+1)} = (1 - \lambda) p_j^{(k)} + \lambda p_{j,\star}^{(k+1)}. \quad (32)$$

Manufactured fixed points are recovered to approximately 10^{-13} in p_1 and 10^{-15} in p_2 . The iteration is therefore algebraically consistent, although convergence from the naive first-order state is slow because the two response spaces are strongly coupled.

In the radius-by-radius q8 test:

- at $R = 0.24M_{\text{tot}}$, the interior error decreases from 3.65×10^{-2} to 2.98×10^{-2} after three sweeps;
- neighboring radii 0.28, 0.34, and $0.40M_{\text{tot}}$ prefer approximately three to five sweeps;
- the surface residual generally reaches its minimum earlier than the interior metric.

Thus surface convergence is not a sufficient stopping criterion. The iteration is currently best interpreted as an early-stopped regularization path, not as an algorithm that should always be run to a fixed point.

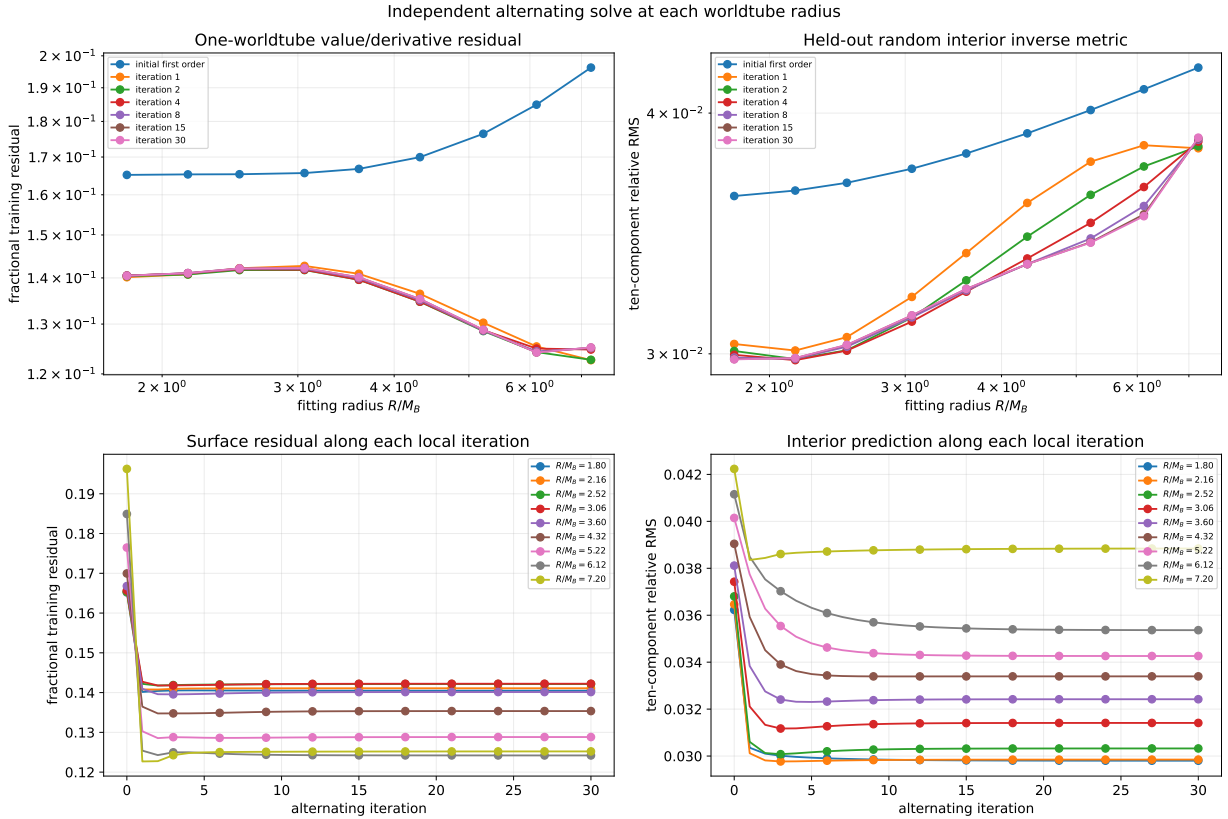


Figure 5: Radius-by-radius Gauss-Seidel iteration. Surface and interior diagnostics generally prefer different stopping times.

16 Time derivatives and evolution consistency

The first-order generalized-harmonic variables provide

$$\partial_T g_{ab} = \beta^i \Phi_{iab} - \alpha \Pi_{ab}, \quad \partial_T G^{ab} = -G^{ac} G^{bd} \partial_T g_{cd}. \quad (33)$$

At fixed local coordinates on a sphere moving with velocity v_{center}^i ,

$$D_T G^{ab} = \partial_T G^{ab} + v_{\text{center}}^i \partial_i G^{ab}. \quad (34)$$

The Π, Φ -constructed derivative agrees with a time derivative of independently interpolated moving-sphere data to approximately 2×10^{-4} at $R = 0.2M_{\text{tot}}$.

Let

$$a = \dot{q}^0, \quad \Lambda_{ij} = L_{ij}/M_B. \quad (35)$$

Since $T = t + q^0(t)$,

$$\ddot{q}^0 = (1 + a) \frac{da}{dT}, \quad \dot{L}_{ij} = M_B(1 + a) \frac{d\Lambda_{ij}}{dT}. \quad (36)$$

At $R = 0.2M_{\text{tot}}$, the directly measured scales are

	Π, Φ direct	finite difference	free second-order fit
RMS $ \ddot{q}^0 $	3.95×10^{-5}	4.42×10^{-5}	4.08×10^{-2}
RMS $ M_B \dot{\Lambda} $	1.25×10^{-5}	1.96×10^{-5}	7.16×10^{-3}

The freely fitted time jets are therefore inflated by roughly three orders of magnitude. They act as effective residual-reduction directions rather than physical derivatives. Moreover, the true vector rates $\dot{b}_i$ and $\ddot{q}^i$ have norms of a few 10^{-3} , larger than the physical clock and strain rates. They cannot be omitted from a dynamical second-order subsystem.

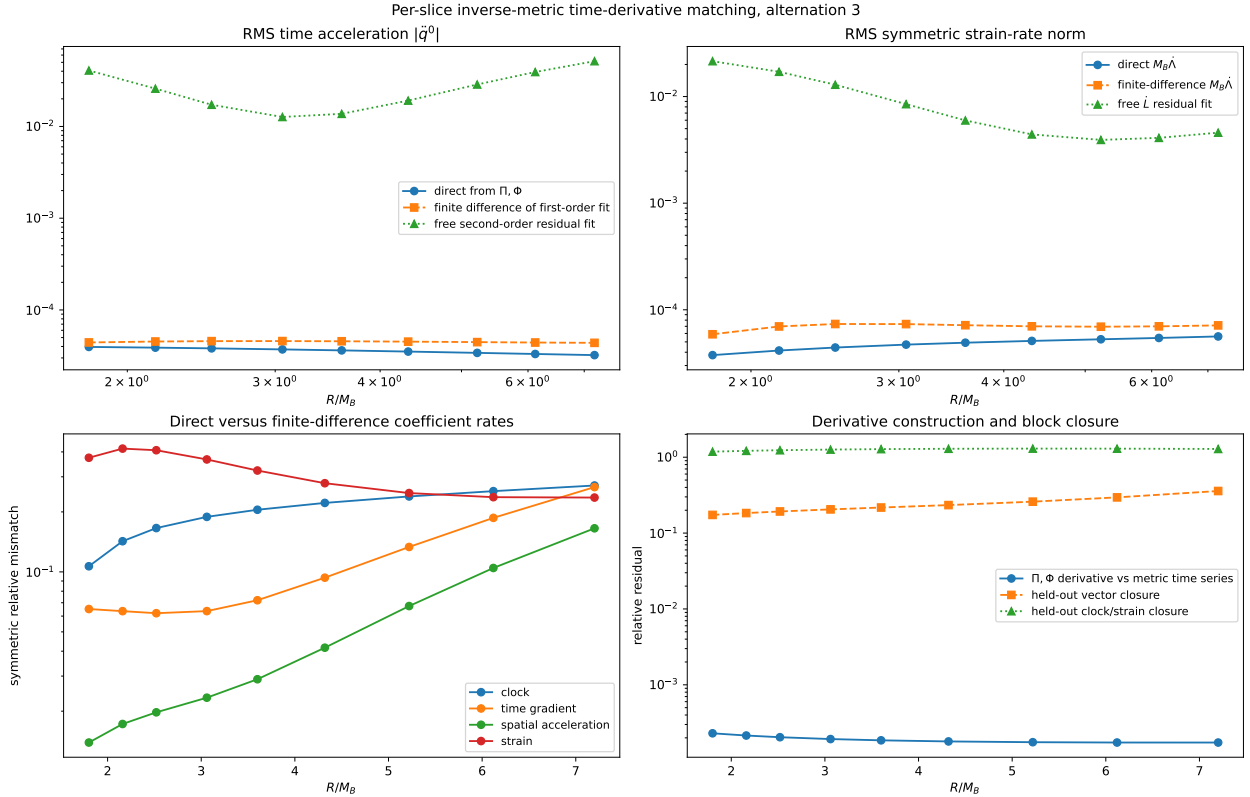


Figure 6: Direct Π, Φ measurements, finite differences of the first-order history, and freely fitted second-order rates.

16.1 First-order evolution and center velocity

Integrating the direct rates from one initial value reproduces the evolution of $\dot{q}^i$ to 1.36% and that of $\ddot{q}^0$ to 10.4% at the smallest radius.

The curvature-center velocity gives the absolute kinematic relation

$$\dot{q}^i = (1 + \dot{q}^0) \frac{dz_{\text{center}}^i}{dT}. \quad (37)$$

It agrees with the algebraic metric fit to 2.3% at $R = 0.2M_{\text{tot}}$ and approximately 2.1% at $R = 0.28M_{\text{tot}}$. Omitting the time-coordinate factor gives an error near 10%.

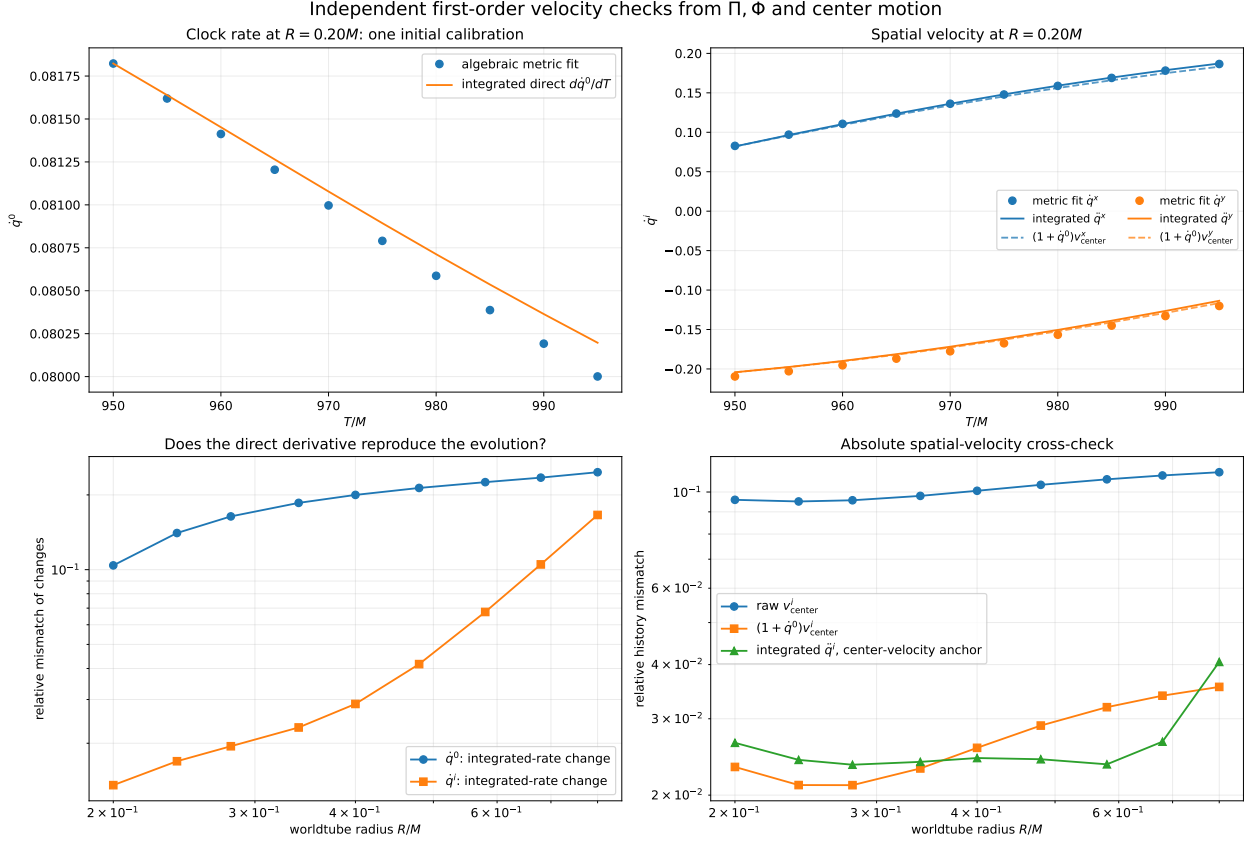


Figure 7: Independent first-order velocity checks from direct metric derivatives and curvature-center motion.

16.2 Metric-only translation velocity

The comoving derivative in [equation \(34\)](#) cancels center translation and measures $\tilde{q}^i$. The fixed-inertial derivative instead satisfies

$$\partial_T G^{ab} = -\partial_i G_{(0)}^{ab} \frac{dq^i}{dT} + \sum_A \mathcal{R}_A^{ab} \frac{dp_A}{dT}. \quad (38)$$

A preliminary full-rank sixteen-parameter fit gives a metric-only center velocity. At $R = 0.2M_{\text{tot}}$, it agrees with the Kretschmann-center velocity to approximately 2.7%, and the corresponding harmonic-time $\tilde{q}^i$ agrees with the algebraic first-order fit to approximately 4.9%.

There is no analogous absolute q^0 translation measurement because the Schwarzschild background is stationary. The derivative determines $\tilde{q}^0$; $\tilde{q}^0$ still requires an algebraic fit or one initial calibration.

17 Limitations

- The q4 $T = 850M_{\text{tot}}$ production slice has no matched fine-resolution copy; the downloaded fine data stop near $T = 700M_{\text{tot}}$.
- The q8 study uses full-resolution output, but $q = 8$ is not yet an asymptotically small mass ratio.
- The q4 small black hole had a residual spin dominated by $\chi_z \simeq -2.23 \times 10^{-4}$. Spin has not been added to the background model.
- The multi-slice analysis spans $45M_{\text{tot}}$. Direct per-slice derivatives remove the main cadence limitation, but longer histories are needed for dynamical stability tests.
- The magnetic response remains the least well verified analytic and numerical sector.
- A small training residual is never sufficient evidence of physical parameter recovery.

18 Current assessment

The analysis supports the following picture:

1. The corrected coordinate-map and response algebra is internally consistent for the implemented families.
2. First order gives a large and robust improvement over centered Schwarzschild.
3. V1 is the strongest current vector estimator, and radial derivatives modestly improve it.
4. C3 is the most conservative current clock/strain estimator, although its closure remains weaker.
5. The complete triangular second-order solve is not physically validated despite passing every manufactured algebraic test.
6. Useful second-order information is concentrated in spatial-strain rate, time acceleration, and the polar- $J = 2$ response space.
7. Radial derivatives add genuine information at both first and second order.
8. Direct time derivatives must replace or constrain freely fitted evolution jets.
9. A few Gauss–Seidel sweeps can decontaminate the two orders, but the iteration requires validation-based early stopping.
10. The fixed-inertial metric derivative supplies an independent measurement of spatial center velocity.

The emerging construction is therefore a hierarchy of physically organized overdetermined systems rather than one unrestricted global least-squares fit. Each block must be judged by radius stability, omitted-block closure, derivative consistency, and held-out interior prediction.

A Compact numerical summary

test	representative value	interpretation
q8 Schwarzschild interior error	2.21×10^{-1}	zeroth-order baseline
q8 V1+C3 interior error	3.61×10^{-2}	factor 6.14 improvement
derivative-assisted first order	3.57×10^{-2}	small vector-sector improvement
seven-parameter second order with radial derivative	$3.17\text{--}3.19 \times 10^{-2}$	conservative useful second-order block
polar- $J = 2$ targeted fit	3.13×10^{-2}	strongest targeted 15-parameter subsystem
globally balanced 43-column fit	3.01×10^{-2}	useful aggregate response, not individual identification
radius-by-radius alternation at $R = 0.24M$	2.98×10^{-2} after 3 sweeps	supports early-stopped decontamination
direct clock acceleration	3.95×10^{-5}	free fit overestimates by about 10^3
direct spatial velocity evolution	1.36% mismatch	strong dynamical consistency
metric-only center velocity	2.7% mismatch	preliminary fixed-inertial derivative result

B Companion artifacts

- `q8_first_order_matching.ipynb`: complete executed single-slice analysis.
- `build_q8_first_order_notebook.py`: reproducible notebook source.
- `temporal_ode_consistency/`: ten-slice and direct Π, Φ derivative analyses.
- `gauge-matching-audit/`: analytic audit, Mathematica notebooks, focused q4 notebooks, and provenance.